

Design Verification Plan and Report (virtual test version)

ED-Compact B-p smartphone cell - case exp/t6_r1_noceiling

VBF-T6R1NOCEILING-DVPR-01

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Design under verification: ED-Compact B-p (Chen2020 NMC811/graphite; compaction + dead-layer thinning + 4 μm negative particles; nominal 5.6 Ah). All results from simulation output files (mechanical values); uncovered conditions honestly N/A.

#	Verification item	Condition	Result value	Determination (vs entry-0 criteria)	Source
1	1C discharge capacity	1C_discharge, SPMe, 25 °C	5.3222 Ah (nominal 5.6 Ah, -5%)	reporting item; nominal reconciliation noted	cell/r2_bp_1c.json:capacity_ah
2	Volumetric energy density	contract caliber (electrolyte excluded)	964.44 Wh/L	PASS (≥ 950)	cell/r2_bp_energy.json:energy_density_wh_l
3	Gravimetric energy density	contract caliber	469.06 Wh/kg	reporting item	cell/r2_bp_energy.json:energy_density_wh_kg
4	Voltage plateau	discharge-time midpoint	3.9214 V	FAIL (< 4.1)	cell/r2_bp_energy.json:midpoint_voltage_v
5	4C fast-charge temperature rise	4C_charge_45C, DFN, lumped thermal, 45 °C ambient, h = 10	T_max = 351.004 K (77.9 °C)	FAIL (> 323.15 K); family datum at h = 25: B 338.7 K - still FAIL	cell/r2_bp_4c.json:T_max_K; cell/r3_bh25_4c.json:T_max_K
6	4C fast-charge plating	same, with plating module	anode potential min = -0.2566 V -> plated = true	FAIL (must be false)	cell/r2_bp_4c.json:anode_potential_v
7	4C charge acceptance	same	0.096 Ah charged at 4C before 4.2 V limit	reporting item (poor fast-charge support, honestly recorded)	cell/r2_bp_4c.json:capacity_ah
8	Anode SEI after 100 cycles	aging_1C_100cyc, SPMe, isothermal, 25 °C	417.74 nm	PASS (≤ 500)	cell/r4_bp_aging.json:sei_thickness_nm_end
9	Voltage window	parameter set	2.5 - 4.2 V	reporting item	Chen2020 parameter set
10	DC resistance	10%-discharge definition	0.364 mohm	reporting item	cell/r2_bp_energy.json:dcr_ohm
11	Nail penetration	-	N/A	N/A (beyond pure-simulation boundary, requires physical experiment)	-
12	Overcharge to thermal runaway	-	N/A	N/A (beyond pure-simulation boundary)	-
13	Crush / drop / vibration	-	N/A	N/A (requires physical experiment)	-
14	Cycle life to 80% capacity	-	Not simulated (aging protocol = 100 cycles; capacity trajectory shows climb-then-saturate artifact - annotated, not treated as normal degradation)	N/A	cell/r4_bp_aging.json:capacity_ah_per_cycle

Conclusion

- PASS items: 2 of 5 acceptance metrics (volumetric ED ≥ 950 Wh/L; SEI ≤ 500 nm after 100 cycles) plus reporting items 1, 3, 9, 10.
- FAIL items: voltage plateau ($3.9214 \text{ V} < 4.1 \text{ V}$), 4C plating (plated = true), 4C/45 °C T_max ($351.0 \text{ K} > 323.15 \text{ K}$ at h = 10; 338.7 K at h = 25).
- Uncovered items: 11-14 (physical-experiment territory), 4C acceptance beyond 0.096 Ah (voltage-limited), B-p SEI at elevated temperature (45 °C aging protocol not run for this candidate).
- Overall: design does not satisfy the full acceptance contract (consistent with log final verdict "not achieved"); the in-boundary achievements and the per-metric relaxation mapping are documented in log entry final.escalation.